

Node.js Lab: EventEmitter Memory Leak Diagnostics

Duration : 60 Min.

Scenario

A server endpoint registers custom event listeners to a global event emitter for each API call but fails to remove them, creating a massive memory leak under concurrent load.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Identify the event listener leak inside `server.js` and modify it to register listeners dynamically using `once()` or clean up properly upon request execution completion.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	File <code>server.js</code> exists in student workspace	2.5 Marks
TC2	Express server compiles and starts successfully	2.5 Marks
TC3	Target route <code>/api/listen</code> returns HTTP 200 OK status	2.5 Marks
TC4	Global emitter listeners do not grow unbounded under load	2.5 Marks

TC5	Correct usage of once() or removeListener() handlers	2.5 Marks
TC6	Listener limit warnings do not trigger	2.5 Marks
TC7	Request event responses are handled correctly	2.5 Marks
TC8	Clean shutdown and listener cleanup verified	2.5 Marks

Total Score: 20 Marks